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MAM1019H

Hand-in 8

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Answers

1. (a) True
(b) False
(c) True
(d) False
2. (a) We have that $[a, b] = [n, 0]$ or $[0, n]$ for some $n \in \mathbb{N}_0$. Now $[n, 0]$ is denoted by n and $[0, n]$ is denoted by $-n$, thus for $[a, b]$ to represent -3 , $[a, b] = [0, 3]$. Similarly for $[c, d]$ to represent 2 , we need $[c, d] = [2, 0]$
(b) $[e, f] = [a, b] +_{\mathbb{Z}} [c, d] = [0, 3] +_{\mathbb{Z}} [2, 0] = [0 + 2, 3 + 0] = [2, 3]$
(c) $[2, 3]$ is represented by $(2, 3)$. Now we need $(2, 3) \sim (0, n)$ or $(2, 3) \sim (n, 0)$. Notice if $(2, 3) \sim (n, 0)$, then $2 + 0 = 3 + n$ but $n \in \mathbb{N}_0$ and $2 < 3$ yielding no solution. So $(2, 3) \sim (0, n) \iff 2 + n = 3 + 0 \iff n = 1$. So $(2, 3) \sim (0, 1)$.
(d) Since $(2, 3) \sim (0, 1)$, we have that $[2, 3] = [0, 1]$ so $[e, f] = [2, 3] = [0, 1]$ is represented as -1
3. $(m, n) \sim (p, 2) \iff m + 2 = p + n$. By commutativity, $p + n = n + p$ so $n + p = m + 2$. Now, by the definition of subtraction, if $n \leq (m + 2)$, then $p = (m + 2) - n$. So we need to show that $n \leq m + 2$. It was given that $n \leq m$, so $n + 2 \leq m + 2$. Also $n \leq n + 2$, so by the transitivity of $\leq$, we have that $n \leq m + 2$. Thus, $p = (m + 2) - n$. We see that $p + n = ((m + 2) - n) + n = m + 2$ so $(m, n) \sim (p, 2)$ so this solution is valid